



FACT SHEET

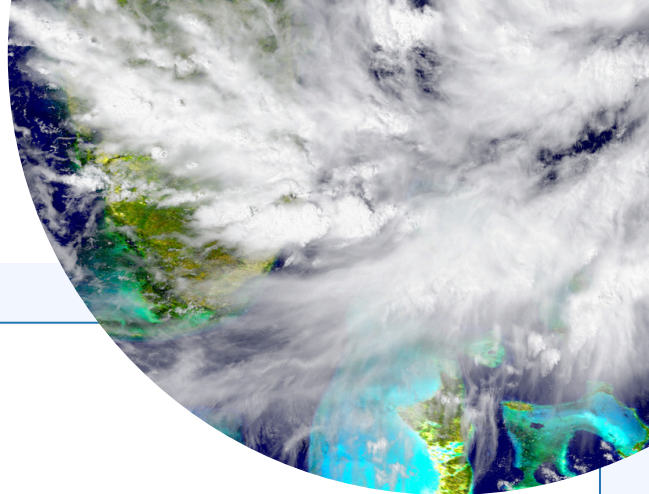


Task 13 Reliability and Performance of Photovoltaic Systems

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Optimisation of Photovoltaic Systems for Different Climates

August
2025



Why must PV systems be optimised for different climates?

Global PV deployment is expanding rapidly into regions with climates once considered too challenging for solar energy: deserts, tropical zones, and snow-rich regions. These environments offer strong PV potential but introduce unique design, operation, and reliability challenges.

The IEA PVPS Task 13 Optimisation of Photovoltaic Systems for Different Climates report provides climate-specific guidance for PV developers, planners, and operators to improve system performance, mitigate degradation, and enhance reliability.

Climate-Specific Stressors

Each climate has its own stressors:



Extreme Temperature Fluctuations



Low Temperatures



Mechanical Stressors
(Snow, Ice, Wind)



UV Exposure



Humidity



High Temperatures



Soiling



Salt Mist

Why Climate-Specific Strategies Matter



Performance Risks:

Locations with high solar potential may experience harsh conditions, which worsen performance losses and increase system stress and degradation risk.



Environmental Stressors:

Temperature extremes, UV irradiation, moisture, snow, sand, and biofilm directly impact energy yield and reliability.



Design Challenges:

Tailored, climate-specific system design and mitigation strategies are essential for maximizing PV lifetime and return on investment.





Cold & Snowy Climates

These climates are found in high-altitude and high-latitude regions with prolonged periods of low temperatures. They feature high seasonal variability in sunlight, with low sun angles at high latitudes or occasional high UV exposure at altitude.

Snow accumulation, ice formation, and freeze-thaw cycles are common, creating challenging conditions for PV system operation and maintenance.

Stressors and Typical Problems

Snow and Ice Formation

Snow accumulation can cause shading, increased soiling losses, and heavy loads that deform module frames or introduce glass breakage. Ice buildup can block light and add mechanical stress, while thawing cycles pose a risk of water ingress and refreezing damage.

Low Temperatures and Thermal Cycling

Cold temperatures can embrittle encapsulants and solder joints, making modules more prone to cracking. Large temperature swings between day and night lead to thermal cycling, causing microcracks and long-term reliability issues.

High Irradiance and UV Exposure

At high altitudes, the increased UV exposure can degrade the backsheets and encapsulants due to high irradiance and snow reflecting 80-90% in the UV range.

Wind and Snow Drifts

Strong winds can cause snow drifts that lead to uneven loading, increasing mechanical stress on modules and structures. Wind can also contribute to cold-related mechanical fatigue in mounting systems.

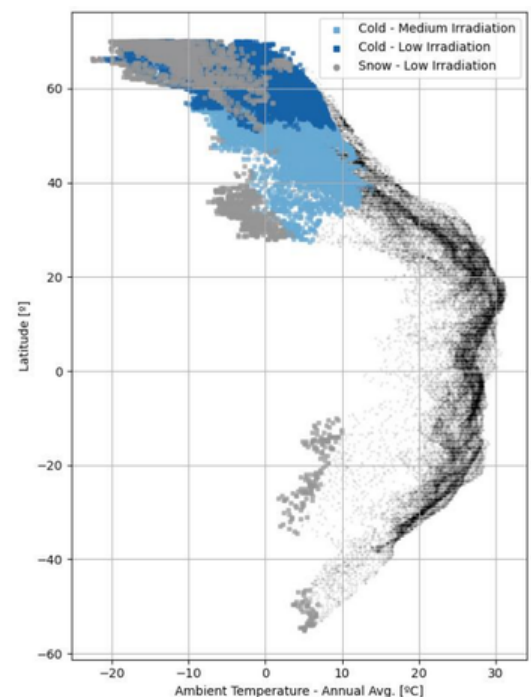


Figure 1: Correlation of latitude with annual ambient temperature highlighting the cold and snowy/polar Köppen Geiger PV (KG PV) climate zones.

Best Practices and Mitigation Strategies



Perform detailed snow load analysis, wind exposure evaluation, and seasonal irradiance studies. Local weather and historical snow data can be used to anticipate accumulation patterns and drifts.



Modules should have reinforced frames, thicker glass, and flexible encapsulants to handle snow loads and cold temperatures. UV-resistant materials are ideal to withstand high-altitude UV exposure.



Use high tilt angles and sufficient ground clearance to reduce snow accumulation, and employ robust racking for snow and wind loads. Shading from snow must be considered when planning cell strings.



☀ Hot & Dry Climates

Extreme heat increases module operating temperatures, causing hot spots and accelerating degradation of encapsulants and backsheets. Dust and sand lead to soiling and abrasion, reducing light transmission and causing additional surface wear.

Stressors and Typical Problems

Soiling

This contamination reduces the energy conversion and can also cause partial shading, leading to permanent damages in case of hot spots.

High Temperatures

Modules and inverters can be severely affected when exposed to high temperatures, leading to lower efficiency and performance losses.

Thermal Cycling

High temperature variations can induce stresses in the structure of PV panels, potentially causing delamination, material fatigue, and breakage.

Salty Mist

In coastal areas, salty mist, combined with fine dust containing a high salt concentration, leads to detrimental corrosive reactions.

UV Irradiance

UV-induced degradation can cause surface cracking and delamination, compromising the module's performance and durability.

Strong Winds

Winds favour the transportation of soiling and cause strong mechanical-load-stress and vibration, potentially leading to damages.

Dust and Sandstorms

Sandstorms lower visibility, reduce the intensity of the surface irradiance and accelerate soiling on PV modules.

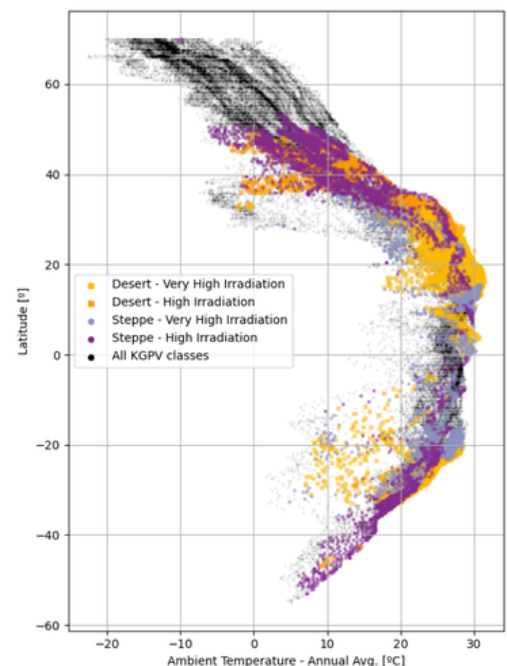


Figure 2: Correlation of latitude with annual ambient temperature highlighting the desert and steppe KGPV climate zones.

Best Practices and Mitigation Strategies



Assess soiling rates, dust composition, wind patterns, and temperature swings using local climate data to plan cleaning needs and optimise system design and O&M strategies for the site.



Modules should use UV- and heat-resistant encapsulants and backsheets, thicker glass for sand protection, and anti-soiling coatings to minimise cleaning frequency and reduce wear.



Design for enhanced ventilation to lower module temperatures, robust structures for wind and sand storms, and optimised tilt and spacing to reduce soiling while maximizing energy capture.



☁ Hot & Humid Climates

High humidity and frequent rain can lead to moisture ingress, corrosion of junction boxes and electrical connections, and delamination. Warm, wet conditions promote biofilm and algae growth on module surfaces, reducing light capture and increasing cleaning needs. Persistent high temperatures can also lead to hot spots and accelerate degradation of PV components.

Stressors and Typical Problems

High Humidity and High Temperatures

High humidity worsens the impact of high temperatures, as heat is not easily dissipated from the modules and water droplets can be accumulated on solar panels. This combination accelerates degradation mechanisms in PV systems.

Enhanced Dust Adhesion

In regions with high humidity, dust particles adhere more easily and strongly to module glass covers, which then require thorough cleaning. Additionally, dew can also cause cementation of dust particles, making them hard to remove by natural cleaning.

Biological Growth

Organic growth is a unique challenge in humid climates, and can negatively impact performance and increase maintenance costs. Beyond the modules, components such as mounting structures, cables, and junction boxes are also susceptible to biofouling.

High Irradiation and Cloud Cover Changes

Tropical and subtropical locations can be characterised by rapid and frequent variations in irradiance, due to frequent cloud cover changes, which cause rapid, low-amplitude temperature changes in PV modules.

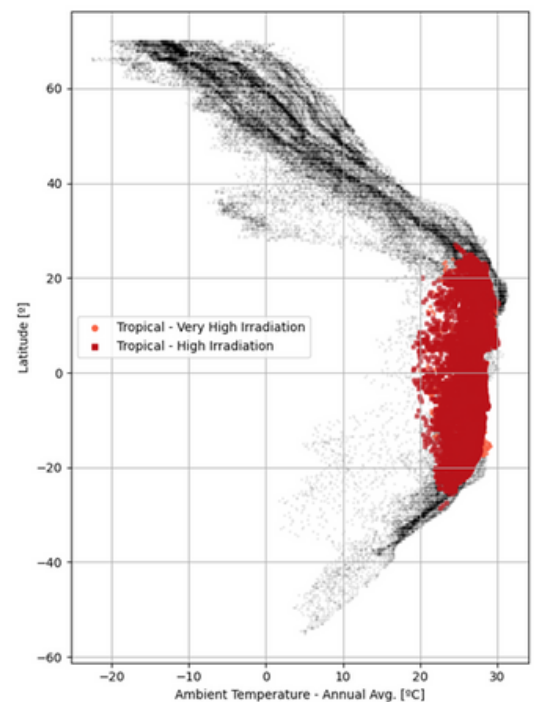


Figure 3: Correlation of latitude with annual ambient temperature highlighting the tropical KGPV climate zones.

Best Practices and Mitigation Strategies



Assess humidity, rainfall intensity, biofilm formation potential, and local irradiance using weather data and field measurements to inform cleaning strategies and site-specific design.



Use moisture-resistant encapsulants, high-barrier backsheets, corrosion-resistant junction boxes, and well-sealed edges to prevent water ingress, delamination, and corrosion in high-humidity climates.



Design for effective drainage and airflow to minimise moisture accumulation, using higher than optimal tilt angles and robust racking to reduce water pooling and biofilm buildup while ensuring easy maintenance.



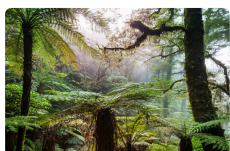
Key Takeaways

Standard modules are insufficient



In cases of limited knowledge about site-specific requirements and/or unavailability of climate-specific PV modules, standard products are often deployed. Furthermore, the push for cheaper modules has driven the trend toward larger PV modules with thinner glass, cheaper encapsulants and backsheets, and reduced frame thickness, which are nowadays increasing the degradation and failure rates in harsh environments.

Climate-specific materials must be deployed



Reinforced front glass, improved frame geometries, micro-crack-resistant interconnections, and advanced encapsulants like POE or silicone enhance durability in harsh conditions. Special coatings for soiling, heat, snow, or corrosion are being explored, but further studies are needed to confirm their longevity and cost-effectiveness.

All solutions require thorough testing and study



Very often, mitigation measures aimed at addressing one issue can inadvertently exacerbate another, making it essential to conduct thorough testing or gain a deeper understanding of actual load conditions to identify the most effective solution.

Further data gathering will prove essential



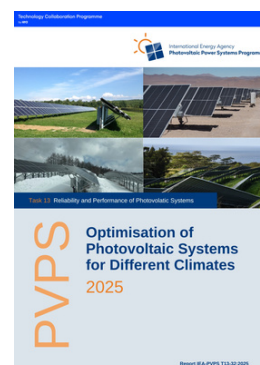
Experience with climate-optimised PV modules is still limited, requiring more field data and lessons learned to be exchanged within the PV community.

Want to know more?

If you are interested in more insights and detailed data, explore the full report:
[“Optimisation of Photovoltaic Systems for Different Climates”](#)

About IEA PVPS Task 13

Task 13 aims to enhance the quality, performance, and reliability of PV modules and systems by summarizing technical aspects, gathering global data, and disseminating results through reports, workshops, webinars, and web content. Task 13's expertise ensures relevant analysis for stakeholders, contributing to technology advancement, risk mitigation, and standardization in PV research and industry.



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PV system located in Albuquerque US, source: Sandia/Laurie Burnham; PV system in northern Chile, source: PVRadars/Thore Müller; PV system located at Pitea Sweden, source: OFI/Gabriele Eder; AES Lawai Solar Project- Kauai in Hawaii, US (Photo by Dennis Schroeder, NREL 58001), source: NREL Image Gallery,